

# Structured summary: | Optimal Allocation of Observations across Run-In, Treatment, and Common Close Phases in Longitudinal Clinical Trials

Ronald G. Thomas, Ph.D.\*

August 15, 2026

## Contents

*This document is a structured, section-by-section summary of the key points argued in the companion manuscript, “| Optimal Allocation of Observations across Run-In, Treatment, and Common Close Phases in Longitudinal Clinical Trials.” It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations, simulation detail, and citations are in the main manuscript.*

---

## 1 Introduction

**Additional run-in and common close observations both improve efficiency but with diminishing or  $J_1$ -dependent returns.**

- Run-in observations appended to a fixed treatment phase improve efficiency with diminishing returns.
- Most of the attainable gain comes from the first two run-in observations; the marginal value falls quickly thereafter.
- Common close gains depend on the number of treatment observations  $J_1$ .

**The efficiency results motivate an allocation question left open by Paper 1.**

- Trials operate under constraints on total visits or total duration.
- Paper 1 characterizes efficiency but not the optimal phase split.
- The question is how to distribute observations across the three phases.

---

\*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965